
PARAMETER-WISE FUNCTIONAL SENSITIVITY FOR SUBNETWORK IDENTIFICATION IN NEURAL NETWORKS

Alan Muriithi
Mathematical Institute
University of Oxford
muriithi@maths.ox.ac.uk

Vedanta Thapar
Mathematical Institute
University of Oxford
vedanta.thapar@maths.ox.ac.uk

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ABSTRACT

Understanding how individual parameters contribute to the mapping learned during neural network training is central to improving interpretability, pruning, and the analysis of learned representations. This is especially pertinent when learning physical systems that obey known symmetries, which should therefore be reflected in the mapping learned by the neural network. We introduce parameter-wise functional sensitivity, a differential framework that quantifies the local effect of perturbing individual parameters on a network’s output. We use this framework to identify subnetworks responsible for symmetry preservation in the learned map. Building on the geometry of the neural tangent space, we show how functional sensitivities can be analysed with respect to continuous physical symmetries by projecting infinitesimal symmetry generators onto the parameter tangent space. This yields parameter-level attribution scores and equivariance diagnostics that reveal how symmetry is distributed across a trained model. We evaluate the framework on Hamiltonian neural networks and multi-layer perceptrons (MLPs) trained on dynamical systems with known rotational and translational symmetries. Our experiments demonstrate that Hamiltonian neural networks learn more concentrated and more equivariant symmetry representations, whose preservation depends on a smaller subset of parameters than in standard MLPs. These results establish functional sensitivity as a principled tool for interpreting neural networks through the geometry of parameter space.

Keywords Functional Sensitivity · Neural Networks · Parameter Sensitivity · Equivariance · Subnetworks

1 Introduction

Neural networks can approximate complex functions by constructing a parameterised map and adjusting a large set of parameters in response to provided data. A fundamental question in the study of such parameterised functions is therefore to identify subsets of parameters that may play a dominant role in the behaviour of the learned map. We approach this problem by defining the *functional sensitivity* of a neural network, which measures the local response of the output map to small perturbations of a parameter. Concretely, consider a map $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, parameterised by a set of weights and biases encoded in θ . The pointwise functional sensitivity of the i -th parameter, θ_i , at an input $x \in \mathcal{X}$ is defined as $S_i^*(x; \theta) := \frac{\partial f_\theta}{\partial \theta_i}(x; \theta) \in \mathbb{R}^{|\mathcal{Y}|}$, and the expected functional sensitivity is given by $S_i(\theta) := \mathbb{E}_{x \sim \mathcal{X}}[S_i^*(x; \theta)]$. Mathematically, this object is related to the well-studied

neural tangent kernel (NTK) [3], which considers the inner product between pointwise sensitivities at different inputs for fixed θ . While the NTK has been used as a powerful tool in the asymptotic analysis of neural networks, particularly to study optimisation properties in the infinite-width limit, the functional sensitivity S_i shifts the focus from comparing the response of the map at different inputs to studying and comparing the roles of individual parameters during training.

We study pointwise functional sensitivity in different regimes to identify specific subsets of parameters that contribute to characteristic properties of the learned map, in particular, symmetries over the input domain. We apply this framework to Hamiltonian neural networks designed to learn Hamiltonian systems from sampled trajectory data.

Maiti *et al.* infer the manifestation of symmetries in the learned function through parameter-space correlators [5]. In contrast, our method analyses the local differential geometry of the learned map rather than statistical correlations in parameter space. Furthermore, our approach provides parameter-level attribution of symmetry generators, enabling the identification of the specific parameters that contribute to a given symmetry.

2 Representing Physical Symmetries in Functional Sensitivity Geometry

Let

$$f_\theta: \mathcal{X} \rightarrow \mathbb{R}^m$$

be a differentiable model with parameters $\theta = (\theta_1, \dots, \theta_p)$. Its functional sensitivities are

$$S_i = \mathbb{E}_{x \sim \text{data}} \left[\frac{\partial f_\theta(x)}{\partial \theta_i} \right], \quad i = 1, \dots, p,$$

and the corresponding functional tangent space is

$$T_\theta = \text{span}\{S_1, \dots, S_p\} = \text{Im}(J_\theta), \quad J_\theta = \frac{\partial f_\theta}{\partial \theta}.$$

Thus, T_θ contains precisely the first-order changes in the learned function that can be produced by infinitesimal parameter perturbations:

$$f_{\theta+\delta\theta} = f_\theta + J_\theta \delta\theta + \mathcal{O}(\|\delta\theta\|^2).$$

Although individual sensitivities depend on the chosen parameterisation, the function-space tangent subspace spanned by these sensitivities is invariant under locally invertible reparametrisations of parameter space.

2.1 Continuous Physical Directions

Let $X_1, \dots, X_k$ denote infinitesimal generators of physically meaningful transformations. These may be generators of a continuous symmetry group, such as translations or rotations, or derivatives with respect to parameters paramtrising deformations of a family of physical systems. Their action on the learned function defines intrinsic functional directions

$$g_a = X_a f_\theta, \quad a = 1, \dots, k,$$

which span the physical generator space

$$\mathcal{G} = \text{span}\{g_1, \dots, g_k\}.$$

The vectors g_a are defined in function space, and the central question is whether these intrinsic directions are contained in T_θ .

2.2 Parameter Attribution

When a generator g is represented approximately in T_θ , its parameter-space realisation is obtained from

$$J_\theta c \approx g.$$

As neural network Jacobians are generally rank-deficient, this representation need not be unique. We use the minimum-norm solution

$$c^* = J_\theta^+ g = \arg \min_c \left\{ \|c\|_2 : c \in \arg \min_z \|J_\theta z - g\|_2 \right\},$$

where J_θ^+ is the Moore–Penrose pseudo-inverse. The coefficient c_i^* quantifies the contribution of parameter θ_i to the least-norm infinitesimal displacement that realises g :

$$J_\theta c^* = \sum_{i=1}^p c_i^* S_i \approx g.$$

Parameter-level quantities may also be aggregated across modules or layers to identify where the represented direction is concentrated.

2.3 Quantifying Equivariance Error

Let $h \in G$, where G represents a symmetry group, act on the input as $x \mapsto h \cdot x$. Let $\rho(h)$ be the corresponding output representation. Exact equivariance requires

$$f_\theta(h \cdot x) = \rho(h) f_\theta(x).$$

Invariance is the special case $\rho(h) = I$.

When a symmetry is exact and its action is independent of θ , differentiating the equivariance relation gives

$$S_i(h \cdot x) = \rho(h) S_i(x)$$

for each parameter sensitivity.

This motivates the per-parameter defect

$$E_i(h) = \mathbb{E}_{x \sim \text{data}} \left[\frac{\|S_i(h \cdot x) - \rho(h) S_i(x)\|}{\|S_i(x)\|} \right].$$

A model may compute an approximately equivariant function while individual parameter perturbations fail to transform equivariantly. Thus E_i tests the local parameter geometry underlying the learned mapping.

2.4 Numerical Stability of Jacobian Decomposition

All function-space quantities are evaluated on a common set of probe points and flattened into vectors in the induced empirical inner-product space. If

$$J_\theta = U \Sigma V^\top$$

is the resulting Jacobian, singular directions below a prescribed relative cut-off are discarded. The retained columns of U form an orthonormal basis for the resolved tangent space, the projector is

$$\Pi_{T_\theta} = U_r U_r^\top,$$

and the truncated pseudo-inverse is

$$J_\theta^+ = V_r \Sigma_r^{-1} U_r^\top.$$

This prevents numerically negligible singular directions from being interpreted as resolved physical structure.

3 Experiment Methodology

3.1 Benchmark Systems

We evaluate the proposed framework on two Hamiltonian systems with known continuous symmetry groups.

Mexican-hat potential. The first benchmark is the isotropic quartic Hamiltonian

$$H(q, p) = \frac{1}{2}\|p\|^2 + \frac{1}{2}\alpha\|q\|^2 + \frac{1}{4}\|q\|^4, \quad q, p \in \mathbb{R}^2,$$

whose equations of motion are

$$\dot{q} = p, \quad \dot{p} = -(\alpha + \|q\|^2)q.$$

This system is exactly equivariant under $SO(2)$ rotations,

$$(q, p) \mapsto (R_\theta q, R_\theta p),$$

with infinitesimal generator Ω ,

$$\Omega = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since the force is radial,

$$F(q) = -(\alpha + \|q\|^2)q,$$

the rotational Lie derivative of the force field vanishes identically,

$$\frac{\partial F}{\partial q}(\Omega q) - \Omega F(q) = 0.$$

Two-body problem. For two particles of unit mass with positions $q_i \in \mathbb{R}^2$ and momenta $p_i \in \mathbb{R}^2$, the Hamiltonian is

$$H(q, p) = \frac{1}{2}(\|p_1\|^2 + \|p_2\|^2) - \frac{\alpha}{\|q_1 - q_2\|}, \quad \alpha > 0,$$

which yields

$$\dot{q}_i = p_i, \quad \dot{p}_1 = -\alpha \frac{q_1 - q_2}{\|q_1 - q_2\|^3}, \quad \dot{p}_2 = -\dot{p}_1.$$

The system is equivariant under the Euclidean group $E(2)$: rotations act as

$$(q_i, p_i) \mapsto (R_\theta q_i, R_\theta p_i),$$

while translations act as

$$(q_i, p_i) \mapsto (q_i + a, p_i), \quad a \in \mathbb{R}^2.$$

For a translation generator

$$\xi_t = \begin{pmatrix} t \\ t \end{pmatrix}, \quad t \in \mathbb{R}^2,$$

the corresponding Lie derivative also vanishes,

$$\frac{\partial F}{\partial q} \xi_t = 0,$$

since translations act trivially on the force vectors and F depends only on the relative displacement $q_1 - q_2$.

The corresponding infinitesimal generators Ω act block-diagonally on the position and momentum components, together with the constant translation generators.

3.2 Architectures

Learning Hamiltonian systems using neural networks has received considerable attention in recent years. Hamiltonian neural networks (HNNs), an early architecture for this problem, were introduced by Greydanus *et al.* [2]. These models parametrise a time-independent Hamiltonian $\mathcal{H} \approx \mathcal{H}_\theta(q, p)$ using an MLP with parameters θ . The model is then trained using the loss

$$\mathcal{L}_{(\mathbf{q}, \mathbf{p})}(\theta) = \left\| \frac{\partial \mathcal{H}_\theta}{\partial \mathbf{p}} - \frac{\partial \mathbf{q}}{\partial t} \right\|_2 + \left\| \frac{\partial \mathcal{H}_\theta}{\partial \mathbf{q}} + \frac{\partial \mathbf{p}}{\partial t} \right\|_2, \quad (1)$$

which enforces Hamilton's equations. As such models require accurate time derivatives in the training data, Chen *et al.* [1] introduced symplectic recurrent neural networks (SRNNs), which use symplectic integrators to predict trajectories from time derivatives obtained from the HNN gradients. This allows training with a loss that depends only on the observed trajectory. Such integrators are *structure-preserving*: they preserve the symplectic structure, and consequently phase-space volume, and typically exhibit bounded long-time energy error. To enable learning across varying physical parameters, Thapar *et al.* [6] introduced adaptable SRNNs (ASRNNs), which incorporate parameter channels into SRNNs and have been shown to learn effectively across parameter space from sparse data.

In this work, we use ASRNNs with a leapfrog integrator [4] as our illustrative architecture (see Section 6.1 for details). As a baseline, we also consider vanilla MLPs coupled with a leapfrog integrator. Specifically, we construct a model $\begin{pmatrix} \dot{\mathbf{q}}_\theta \\ \dot{\mathbf{p}}_\theta \end{pmatrix} = f_\theta(\mathbf{q}, \mathbf{p}; \alpha)$, where f_θ is an MLP with parameter set θ . The time derivatives are fed into a leapfrog-type integrator,

$$\mathbf{p}\left(t + \frac{\Delta t}{2}\right) = \mathbf{p}(t) + \frac{\Delta t}{2} f_{\theta,2}(\mathbf{q}(t), \mathbf{p}(t)) \quad (2)$$

$$\mathbf{q}(t + \Delta t) = \mathbf{q}(t) + \Delta t f_{\theta,1}\left(\mathbf{q}(t), \mathbf{p}\left(t + \frac{\Delta t}{2}\right)\right) \quad (3)$$

$$\begin{aligned} \mathbf{p}(t + \Delta t) &= \mathbf{p}\left(t + \frac{\Delta t}{2}\right) \\ &+ \frac{\Delta t}{2} f_{\theta,2}\left(\mathbf{q}(t + \Delta t), \mathbf{p}\left(t + \frac{\Delta t}{2}\right)\right), \end{aligned} \quad (4)$$

where Δt is the chosen time step, $(\dot{\mathbf{q}}_\theta, \dot{\mathbf{p}}_\theta) = (f_{\theta,1}, f_{\theta,2})$, and the dependence of f_θ on α has been suppressed for clarity. Training minimizes the trajectory-matching loss,

$$\mathcal{L}_{(\mathbf{q}_0, \mathbf{p}_0; \alpha)}(\theta) = \sum_{t \in \mathcal{T}} (\|\mathbf{q}(t) - \hat{\mathbf{q}}(t)\|_2 + \|\mathbf{p}(t) - \hat{\mathbf{p}}(t)\|_2), \quad (5)$$

where $(\mathbf{q}_0, \mathbf{p}_0; \alpha)$ specifies the initial conditions and physical parameters, $(\mathbf{q}(t), \mathbf{p}(t))$ and $(\hat{\mathbf{q}}(t), \hat{\mathbf{p}}(t))$ denote the model predictions and ground-truth values at time t , respectively, and $\mathcal{T}$ is the set of observed time points. The leapfrog integrator is used to maintain consistency with the recurrent structure of the ASRNN. This baseline does not inherit the structure-preserving guarantees of a symplectic integrator however, since the MLP outputs are not constrained to define a Hamiltonian vector field. In contrast, in the ASRNN, the time derivatives used for integration are obtained by automatic differentiation of the parametrised Hamiltonian.

The recurrent MLP architecture incorporates no explicit physical structure, whereas the ASRNN parametrises a Hamiltonian and enforces Hamilton’s equations. Although the latter is therefore designed to exhibit Hamiltonian structure and favourable energy behaviour, neither architecture has additional equivariances built in. Consequently, properties such as rotational or translational symmetry must be learned from data.

In what follows, we refer to the recurrent MLP-leapfrog architecture simply as the MLP.

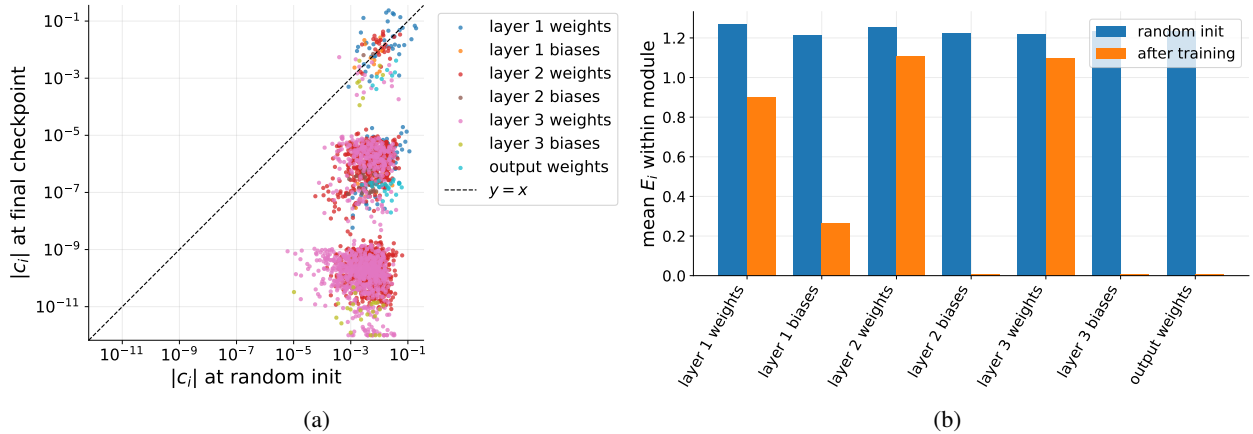


Figure 1: Results for the Mexican hat potential. Panel (a) shows the per-parameter rotation-generator attribution magnitude, $|c_i|$, at random initialisation and after training for the Mexican hat potential learned by an ASRNN. Each point corresponds to a trainable parameter and is coloured by network module. Panel (b) shows the mean sensitivity-equivariance error, E_i , for each network module at random initialisation and after training. Lower values indicate more equivariant parameter sensitivities.

Figure 1a shows that a small subset of parameters have attribution coefficients one to two orders of magnitude above the majority of parameters, indicating that this subset of parameters is significantly more important in realising the symmetry generator in comparison to the rest of the network. Consequently, perturbations to these parameters are expected to contribute more strongly to symmetry breaking in the learned potential. For a fully equivariant network, $|c_i| = 0$ for all θ_i . Figure 1b aggregates the per-parameter sensitivity-equivariance error, E_i , by network module. Here, E_i measures the relative error with which

4 Numerical Results

Figure 1a compares the per-parameter rotation-generator attribution magnitudes, $|c_i|$, at random initialisation and after training. The attribution coefficients are obtained by projecting the infinitesimal rotation generator onto the network’s functional tangent space, thereby providing a quantitative measure of each parameter’s contribution to the representation of rotational symmetry. The scatter plot therefore reveals how training redistributes the representation of rotational symmetry across the parameter space. Parameters lying close to the diagonal retain similar attribution throughout optimisation, whereas deviations from the diagonal indicate parameters whose contribution to the learned symmetry has either increased or decreased. The ASRNN assumes a separable Hamiltonian and uses independent networks for the potential ($\mathcal{V}_{\theta_2}$) and kinetic ($\mathcal{K}_{\theta_1}$) energies; hence, equivariance of the potential is realised solely by $\mathcal{V}_{\theta_2}$. Thus, the layer-wise parameters shown in all figures refer to θ_2 .

each parameter’s functional sensitivity transforms under finite rotations, so smaller values correspond to more equivariant sensitivities. Comparing the randomly initialised and trained models identifies the modules that contribute most to the improvement in rotational equivariance during optimisation and those that remain the dominant sources of equivariance error after training.

Figure 2 shows the parameter attribution coefficients for the rotational and translational generators in the two-body problem. As in Figure 1a, the attribution magnitudes span

several orders of magnitude, although their grouping by magnitude is less distinct. A small subset of parameters exhibits increased attribution after training however, and retains substantially larger attribution magnitudes than the majority of parameters.

The force field learned by the ASRNN for the two-body problem (Figure 7) has a higher equivariance error than the learned field for the Mexican hat potential (Figure 3). The learned function is therefore not fully equivariant, and a clearer separation of parameter attributions may emerge as the equivariance error decreases.

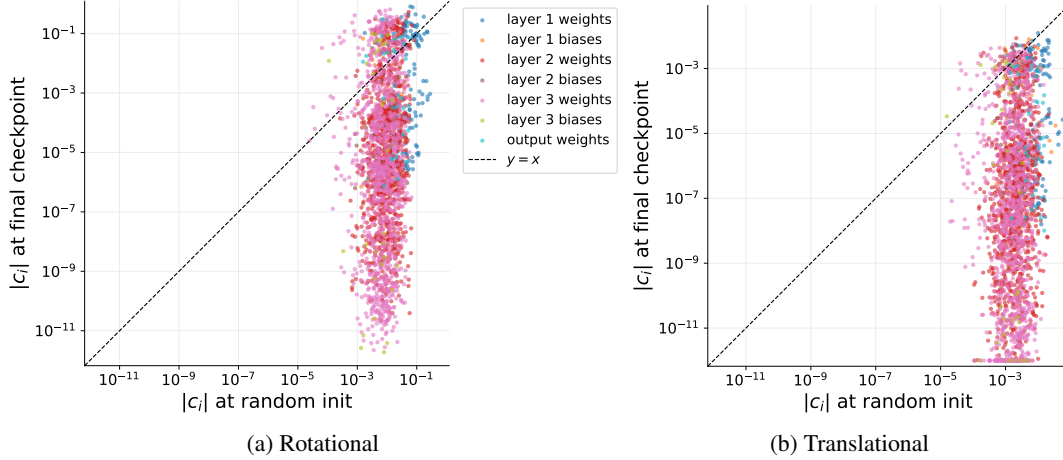


Figure 2: Per-parameter symmetry-generator attribution magnitude $|c_i|$ at random initialisation and after training for the two-body problem learned by an ASRNN. Each point corresponds to a trainable parameter and is coloured by network module.

The ASRNN learns functional representations with both lower prediction error and lower equivariance residuals than the MLP. Parameter-wise symmetry-attribution and module-wise equivariance-error plots for the MLP are provided in the Supplementary Material (Section 6), together with plots of the learned fields and training-loss curves for the Mexican hat benchmark.

5 Discussion

The results of Section 4 indicate that neural networks can learn representations that reflect the symmetries of a physical system, and that these symmetries are manifested in the tangent space of the learned function.

We observe that the ASRNN learns functional representations that better respect the underlying symmetries of the respective systems than those learned by the MLP. Symmetry attribution is also more concentrated among a subset of parameters in the ASRNN than in the MLP. Symmetry attribution is also concentrated among fewer parameters in the ASRNN than in the MLP. This behaviour may be attributable, in part, to the Hamiltonian structure explicitly imposed by the ASRNN, which provides an inductive bias consistent with the underlying physical dynamics.

The concentration of symmetry attribution among a subset of parameters suggests that symmetry need not be represented uniformly across the parameter space. Rather, dif-

ferent parameters can contribute substantially differently to the representation of the same functional symmetry. This observation highlights a distinction between function-level equivariance and its parameter-level realisation: two networks may exhibit similar symmetries at the function level while encoding those symmetries differently within their respective tangent spaces.

Our empirical evaluation is restricted to a small set of dynamical systems with analytically known continuous symmetries. Although these systems provide a controlled setting for validating the proposed functional sensitivity framework, they do not establish whether the method generalises to more complex, higher-dimensional physical systems or to settings in which the underlying symmetries are approximate or unknown.

Furthermore, our experiments compare only two model classes, Hamiltonian neural networks and MLPs of modest size, leaving open the question of how the framework behaves across a broader range of architectures, such as graph neural networks, transformers, and convolutional networks, and at substantially larger parameter scales.

Avenues for future work include investigating the mechanisms by which symmetry representations emerge and evolve during training, and determining whether these dynamics are related to changes in function approximation and equivariance error. A further direction is to identify symmetries in the functional tangent space that are not

known a priori, extending the framework from the attribution of known symmetries towards symmetry discovery. Finally, evaluating the framework across a broader range of neural network architectures and higher-dimensional systems would establish its generality and practical scalability.

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6 Supplementary Material

6.1 ASRNN Architecture

ASRNNs parametrise a time-independent separable Hamiltonian using separate neural networks for the kinetic and potential energies, respectively. Specifically, the Hamiltonian is parametrised as $\mathcal{H}(\mathbf{q}, \mathbf{p}; \alpha) \approx \mathcal{K}_{\theta_1}(\mathbf{p}) + \mathcal{V}_{\theta_2}(\mathbf{q}; \alpha)$, where each network is a simple MLP. Recurrence is introduced through a leapfrog integrator. Given $(\mathbf{q}_t, \mathbf{p}_t)$, the next time step is predicted as

$$\mathbf{p}\left(t + \frac{\Delta t}{2}\right) = \mathbf{p}(t) - \frac{\Delta t}{2} \frac{\partial \mathcal{V}_{\theta_2}}{\partial \mathbf{q}} \Big|_t \quad (6)$$

$$\mathbf{q}(t + \Delta t) = \mathbf{q}(t) + \Delta t \frac{\partial \mathcal{K}_{\theta_1}}{\partial \mathbf{p}} \Big|_{t + \frac{\Delta t}{2}} \quad (7)$$

$$\mathbf{p}(t + \Delta t) = \mathbf{p}\left(t + \frac{\Delta t}{2}\right) - \frac{\Delta t}{2} \frac{\partial \mathcal{V}_{\theta_2}}{\partial \mathbf{q}} \Big|_{t + \Delta t}, \quad (8)$$

where Δt is a chosen time step. The model is then trained using the loss defined in Eq. 5. In our experiments, the potential- and kinetic-energy networks were implemented as MLPs with three hidden layers of width 32, following the hyperparameter regime used by Thapar *et al.* [6]. We additionally used L_1 regularisation to encourage sparser parameter representations.

6.2 Mexican Hat Dataset

For the Mexican hat system, Figure 3 compares the learned potential field with the true potential for representative values of the bifurcation parameter α . Since the dynamics depend only on the gradient of the potential, the learned potential is defined only up to an additive constant; consequently, both potentials are shifted to a common minimum before comparison. The close agreement between the learned and analytic surfaces demonstrates that the Hamiltonian network successfully recovers the underlying energy landscape throughout the parameter space, including across the bifurcation at $\alpha = 0$.

Figure 4 shows the training and validation losses throughout optimisation of the ASRNN and MLP on the Mexican hat dataset. The steady reduction in both losses, together with the absence of a widening gap between the training and validation curves, indicates that both networks fit the Mexican hat dynamics without substantial overfitting to the training trajectories. The ASRNN achieves a loss approximately half an order of magnitude lower than that of the MLP.

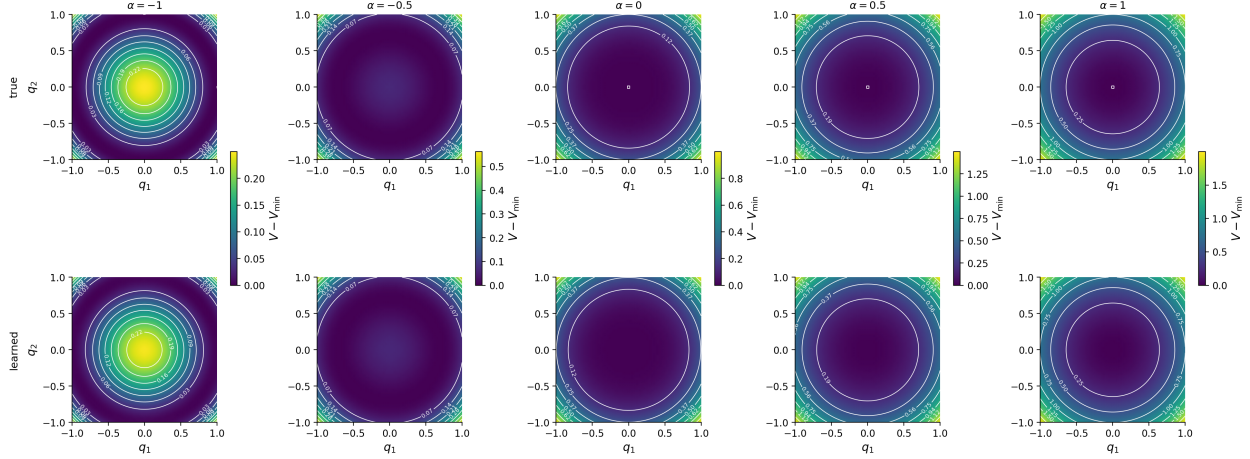


Figure 3: True and learned potential-energy surfaces for representative values of α for the ASRNN. The learned potential field is shown after fixing the arbitrary additive constant by subtracting its minimum value.

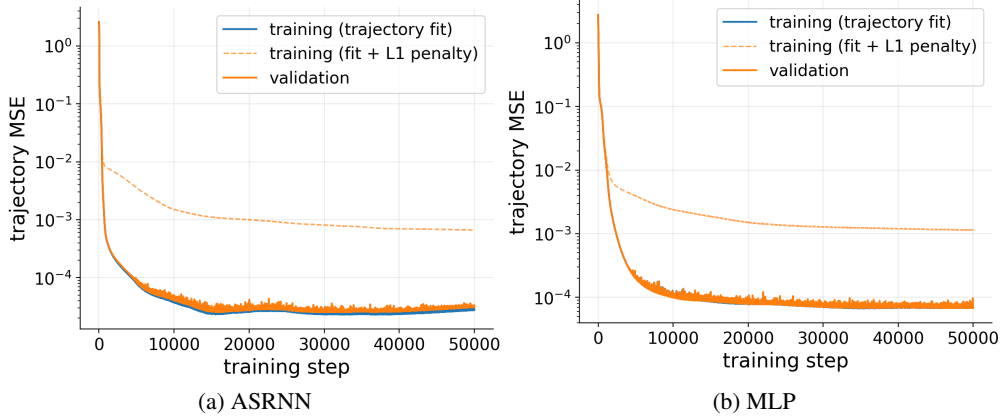


Figure 4: Training and validation losses as a function of optimisation step on the Mexican hat dataset. The validation loss is evaluated on a held-out test set to monitor out of sample error throughout training.

Figure 5(a) shows less distinct separation of the attribution coefficients in the MLP than in the corresponding plot for the ASRNN (Figure 1a).

Figure 5(b) shows that the mean sensitivity-equivariance error of each MLP module changes little from its value at initialisation and remains substantially higher than that of the corresponding ASRNN modules (Figure 1b).

For the Mexican hat system, Figure 6 compares the force field learned by the MLP with the true force field for representative values of the bifurcation parameter α . Unlike the ASRNN, the MLP does not explicitly learn a potential; consequently, only the force field is shown. The learned force field is in good agreement with the true field.

6.3 Two-Body Problem

For the two-body system, Figure 7 compares the learned force field with the true force for representative values of the coupling strength α for the ASRNN. The learned force field is in good agreement with the true force field.

Figure 8 shows the module-wise sensitivity-equivariance errors of the ASRNN for the rotational and translational symmetry generators.

Figure 9 shows $|c_i|$ for the rotational and translational generators before and after training. Notably, the same parameters exhibit large attribution magnitudes for both independent symmetries, indicating that perturbations to these parameters strongly affect both rotational and translational equivariance.

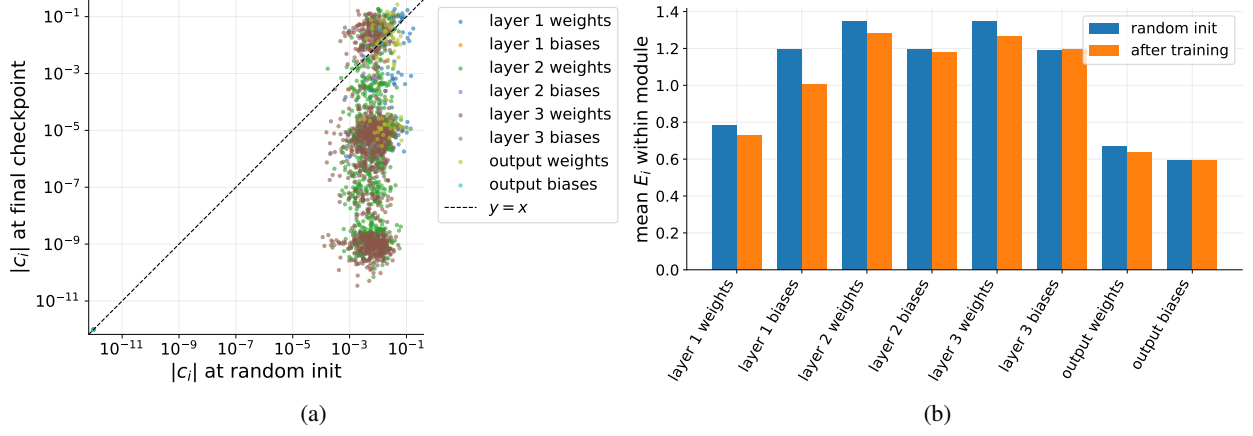


Figure 5: Results for the Mexican hat potential. Panel (a) shows the per-parameter rotation generator attribution magnitude $|c_i|$ at random initialisation and after training for the Mexican hat system learned using an MLP. Each point corresponds to a trainable parameter and is coloured by network module. Panel (b) shows the mean sensitivity-equivariance error E_i for each network module at random initialisation and after training. Lower values indicate more equivariant parameter sensitivities.

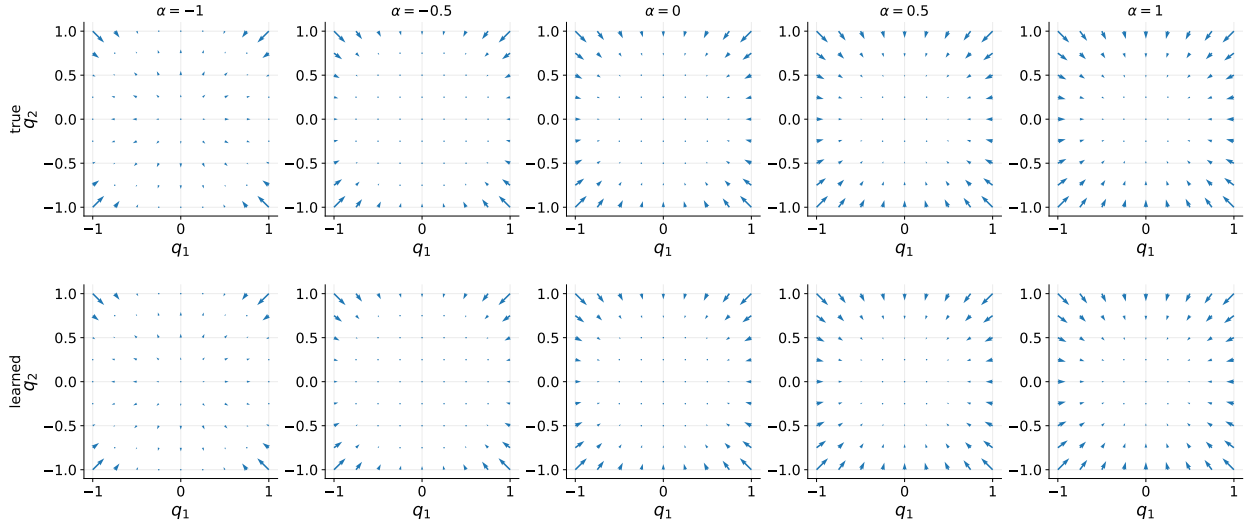


Figure 6: True and learned force vectors for representative values of α for an MLP. The learned force field is shown after fixing the arbitrary additive constant by subtracting its minimum value.

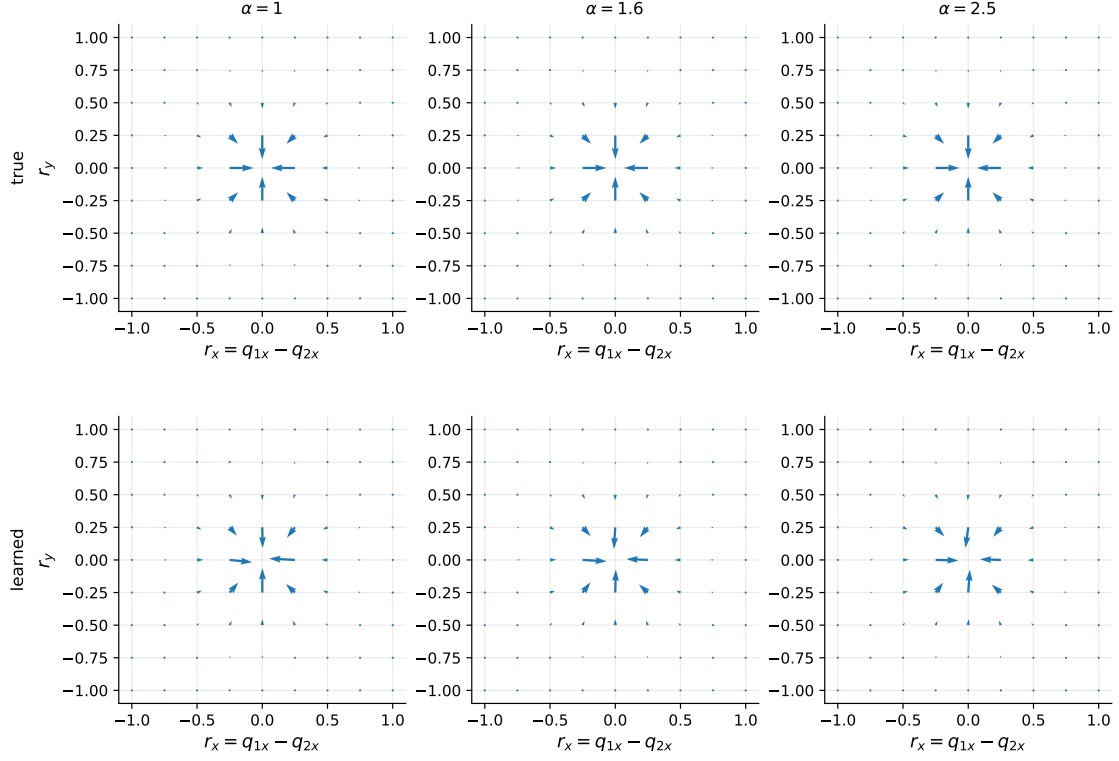


Figure 7: True and learned force vectors for representative values of α for an ASRNN. The learned force field is shown after fixing the arbitrary additive constant by subtracting its minimum value.

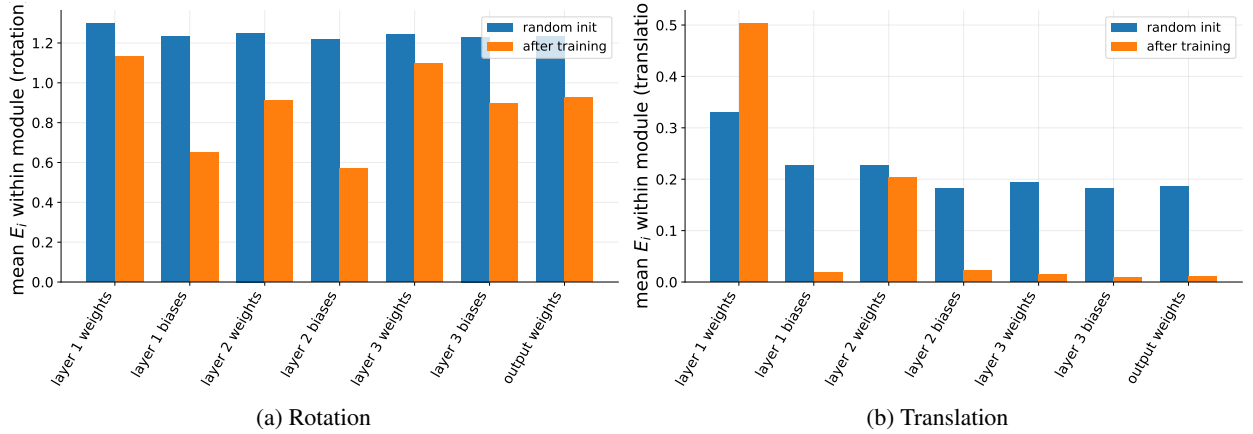


Figure 8: Mean symmetry sensitivity-equivariance errors, E_i , for each network module at random initialisation and after training. Lower values indicate more equivariant parameter sensitivities.

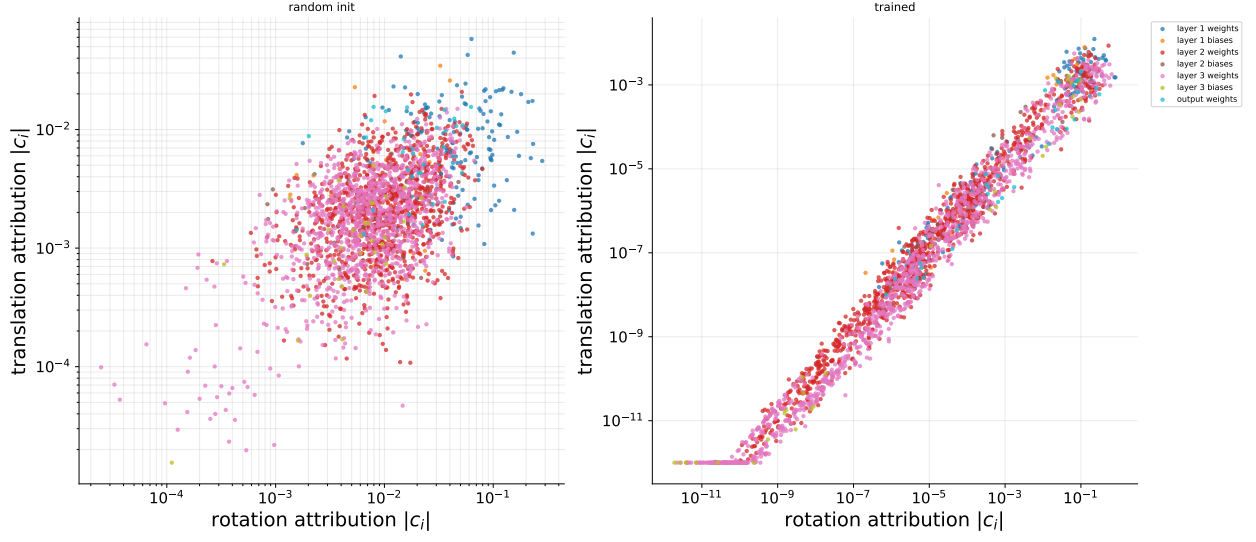


Figure 9: Per-parameter symmetry-generator attribution magnitude $|c_i|$ at random initialisation (left) and after training (right) for the two-body problem with an ASRNN. Each point corresponds to a trainable parameter and is coloured by network module. Rotational c_i values are plotted against the corresponding translational c_i values.

Under the same architectural and data constraints as the ASRNN, the MLP does not adequately learn the two-body dynamics. We therefore omit the corresponding diagnostic plots for this model. This poorer performance is consistent with the findings of Greydanus *et al.* [2], who found that, for near-circular orbits, Hamiltonian neural networks achieve lower training errors than conventional MLPs and more accurately reproduce the resulting trajectories. In contrast, we consider arbitrary closed orbits. The training and validation trajectories used for the two-body problem are shown in Figure 10.

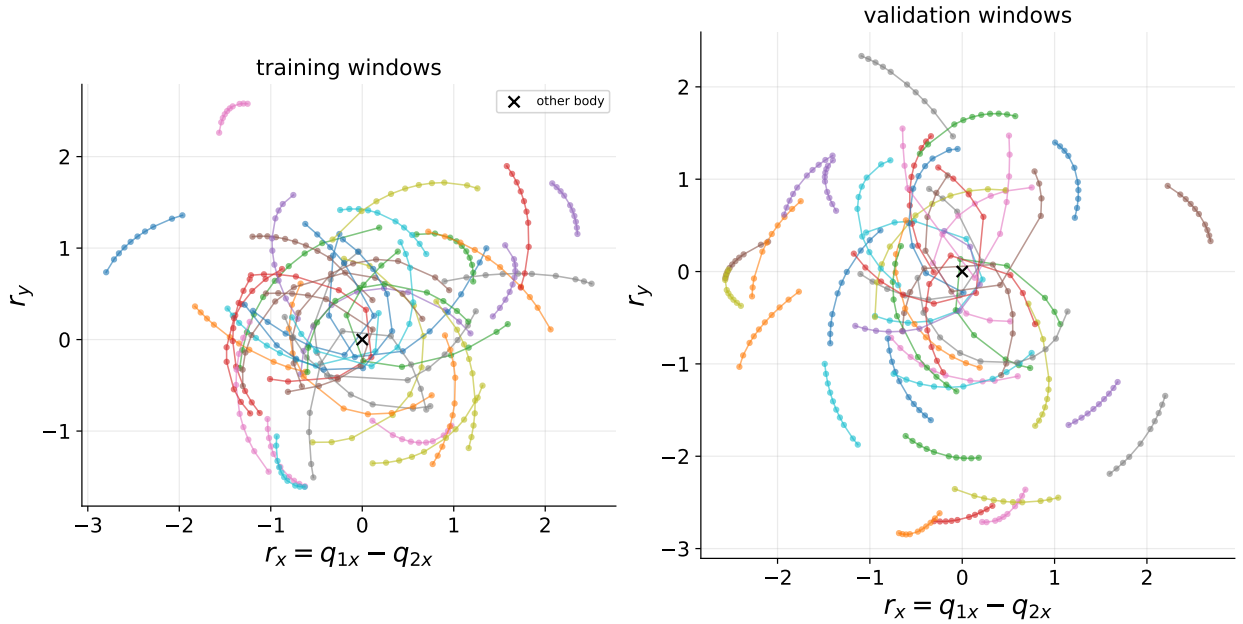


Figure 10: Training and validation trajectories used to train the networks on the two-body problem.